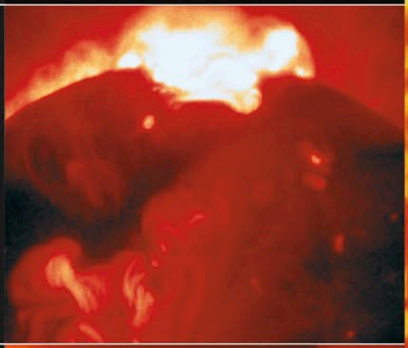
The Sun

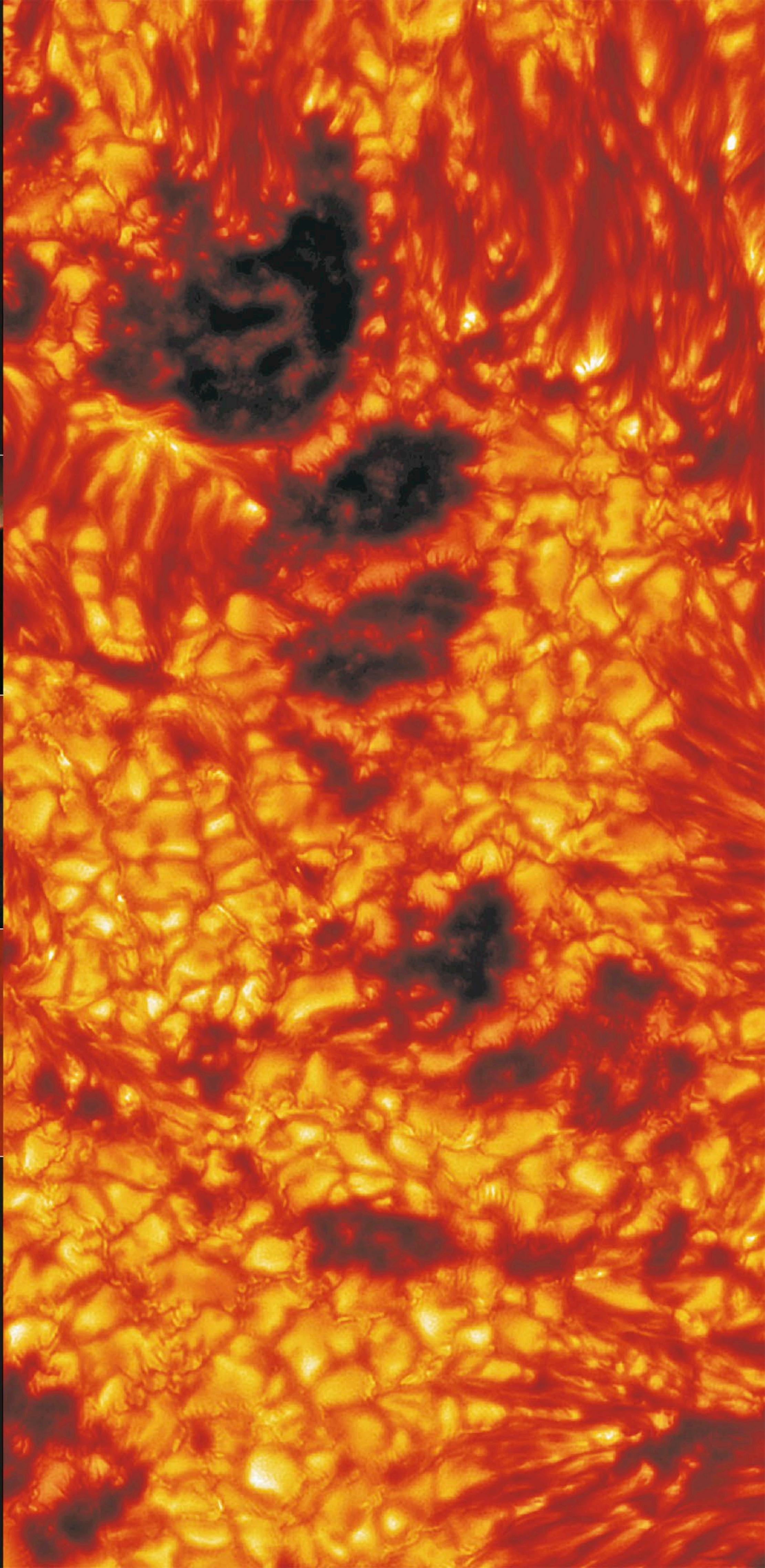
OUR LOCAL STAR
The Sun dominates the solar system. Our chief source of heat and light, it also holds Earth and the rest of the planets in their orbits. This ultraviolet image reveals the dynamic activity in the ultra-hot corona above the Sun's visible surface.

A SOLAR FLARE
The Sun usually appears to the unaided eye as a bright but featureless disk. However, during a total solar eclipse, when light from the disk is blocked out by the Moon, violent flares in the outer layers of the atmosphere can be seen more clearly.



ULTRA-HOT CORONA
The gas in the Sun's corona is heated to several million degrees, causing it to emit X-rays, which show up in this image taken by the Japanese YOHKOH satellite. The dark areas are regions of low-density gas that emit a stream of particles, known as the solar wind, into space.

PROMINENCES
In the corona, electrified gas called plasma forms into huge clouds known as prominences, flowing through the Sun's magnetic field. As the prominence in this image erupts, it hurls plasma out of the Sun's atmosphere and into space.



SUNSPOTS ON THE SOLAR SURFACE
8 light-minutes from Earth
These regions, cooler and darker than the rest of the Sun's surface, are sustained by strong magnetic fields. Some sunspots are large enough to engulf the Earth. Sunspot numbers vary in cycles that take about 11 years to complete, and peaks in the cycle coincide with disturbances, such as aurorae, in our own atmosphere.